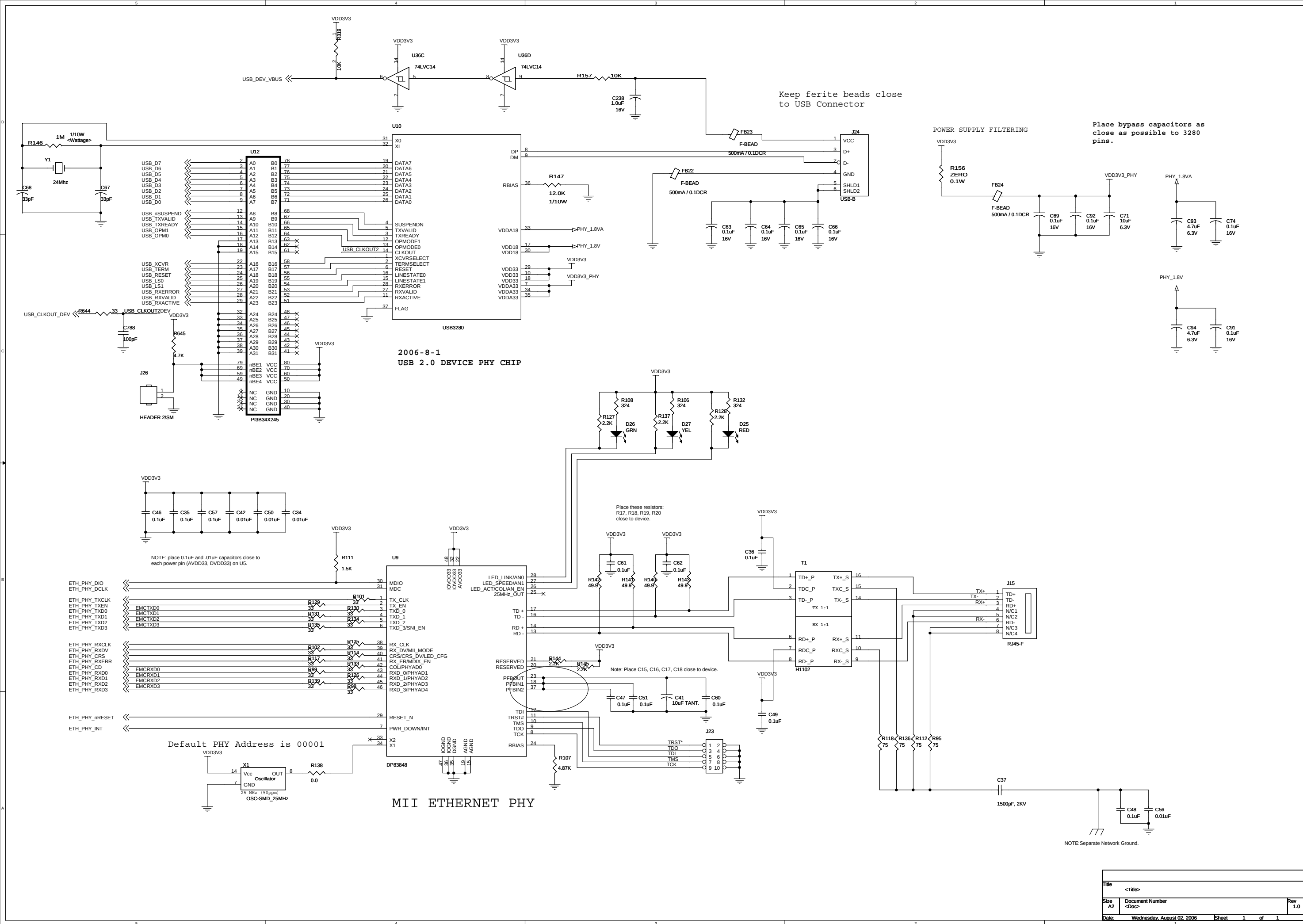
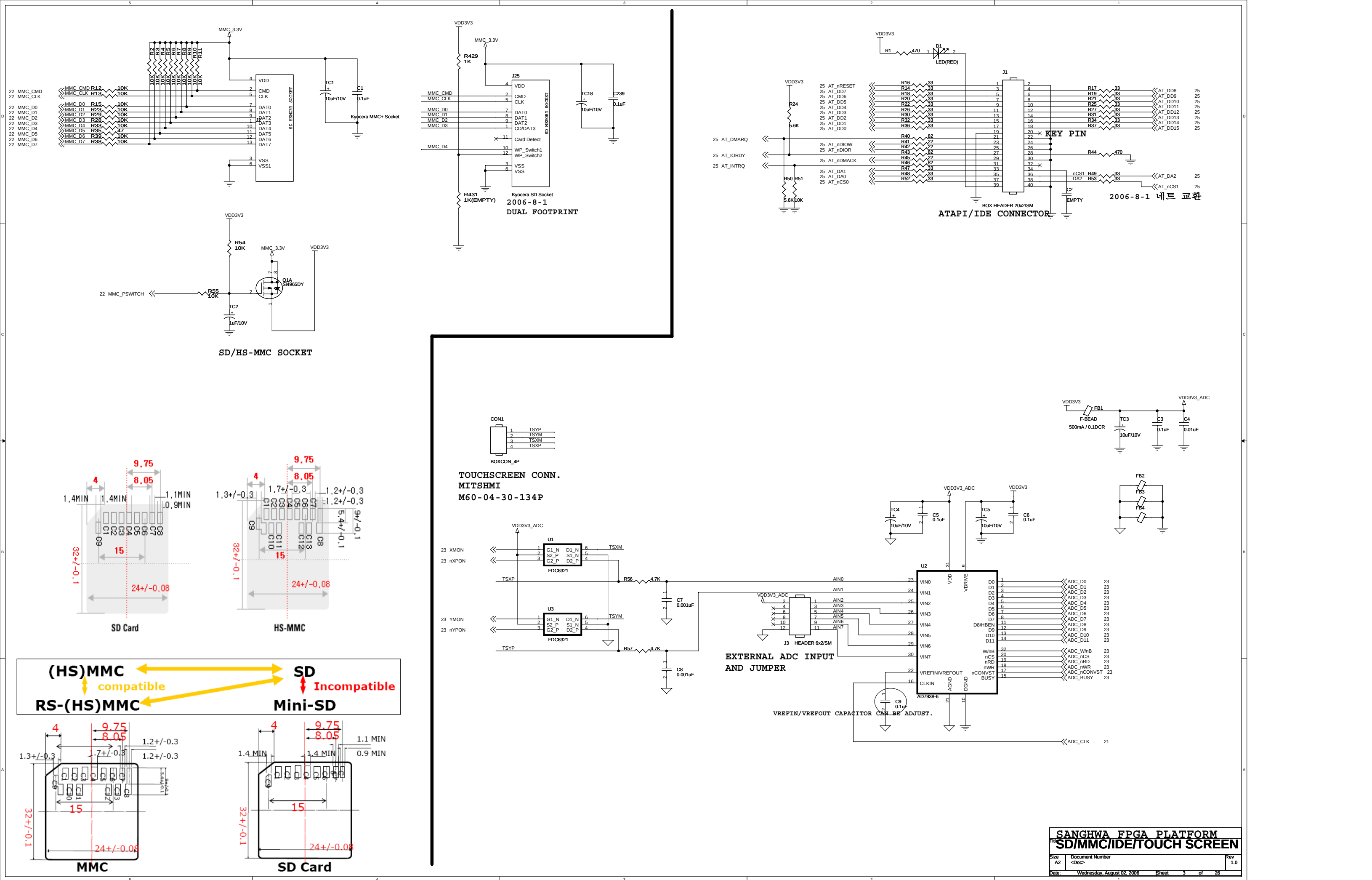
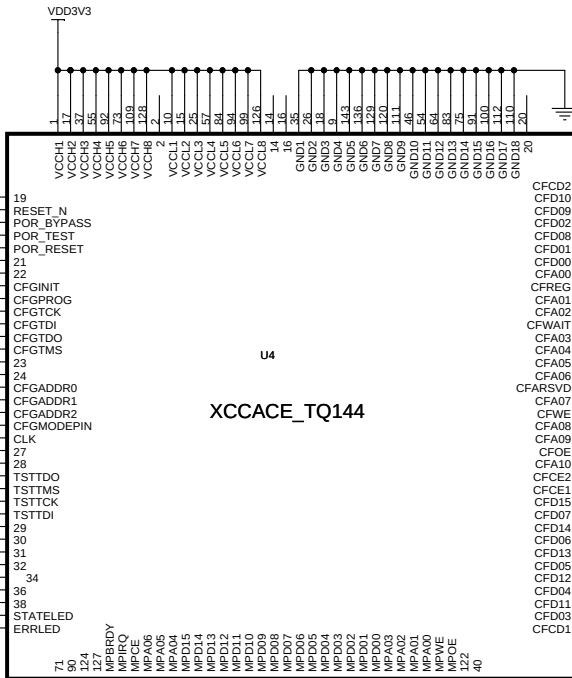
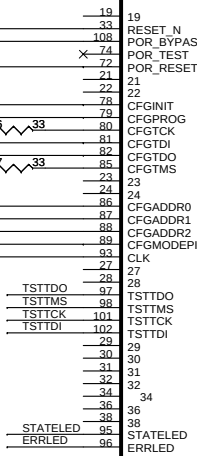
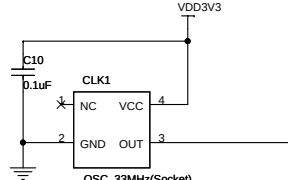
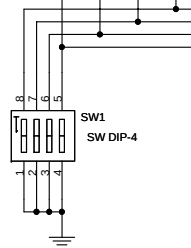
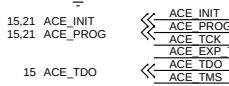
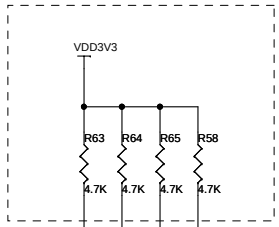
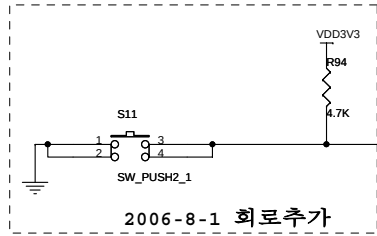
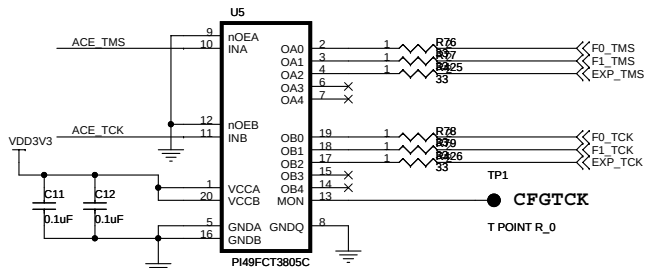
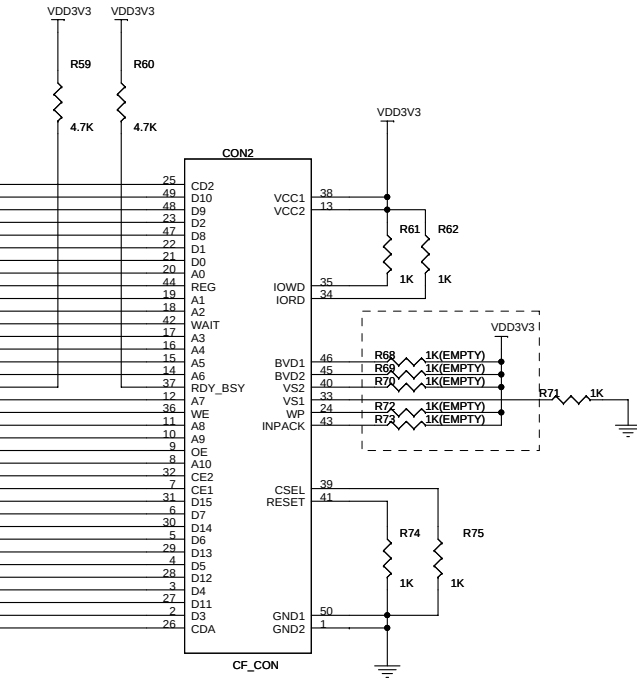


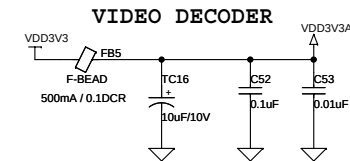
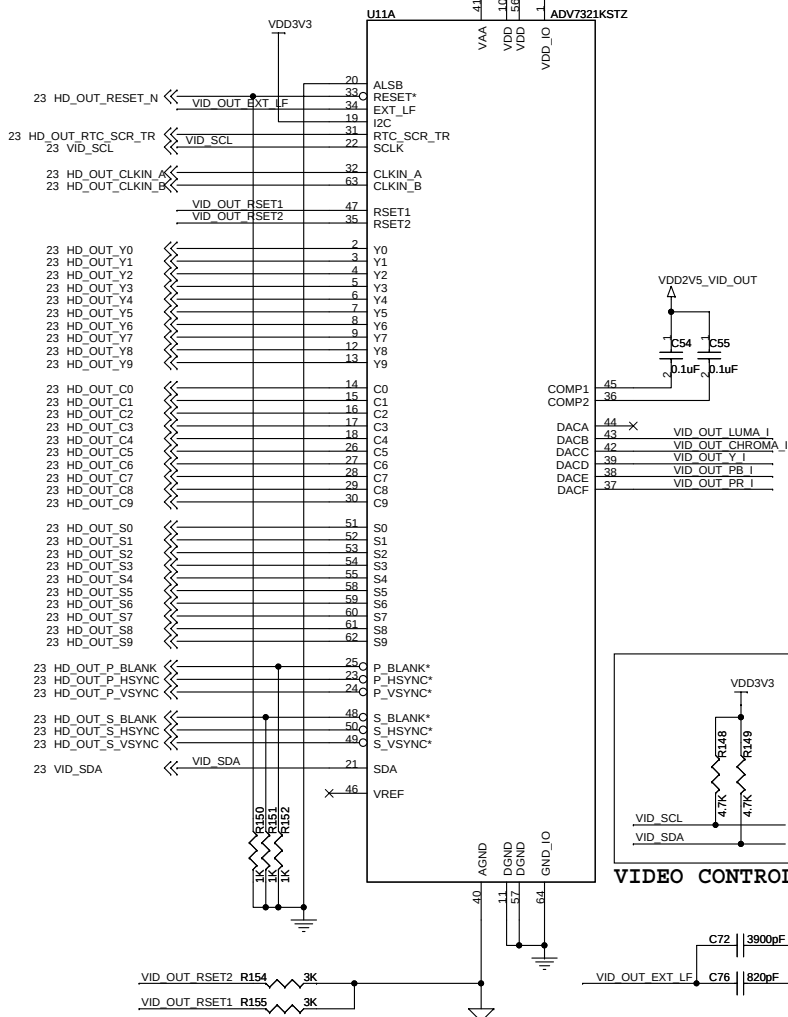
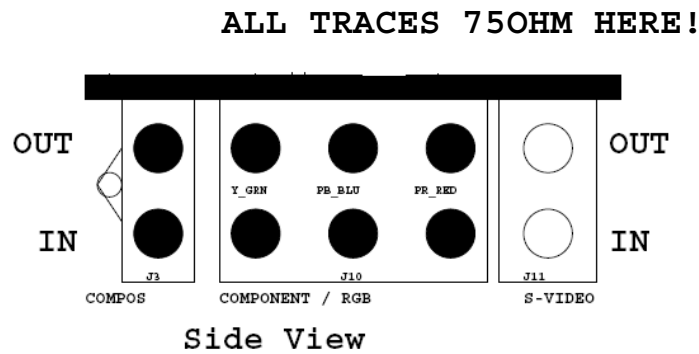
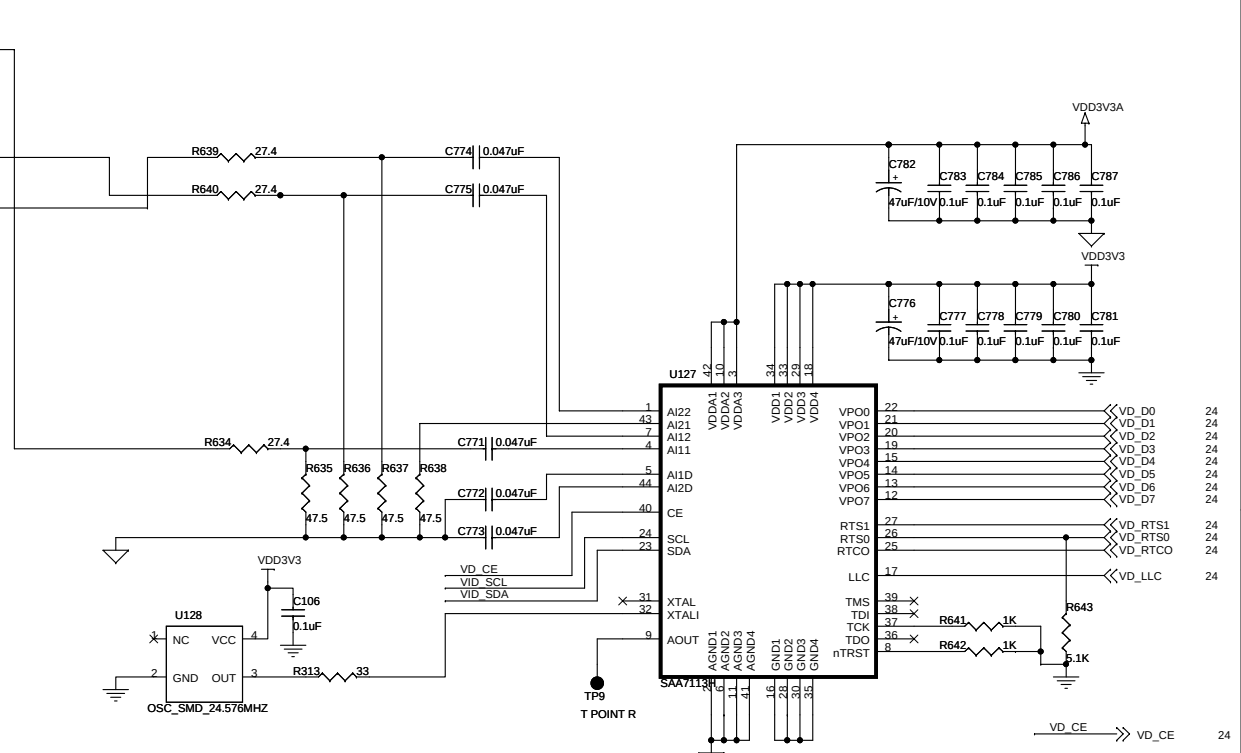
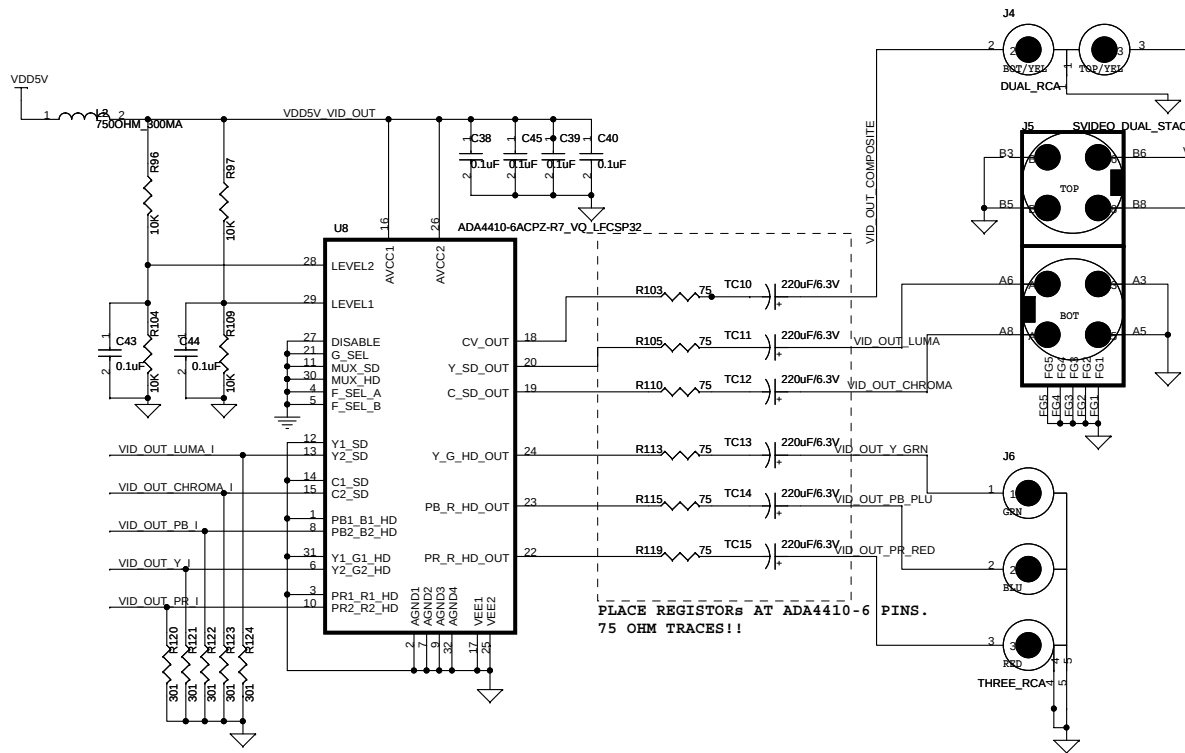




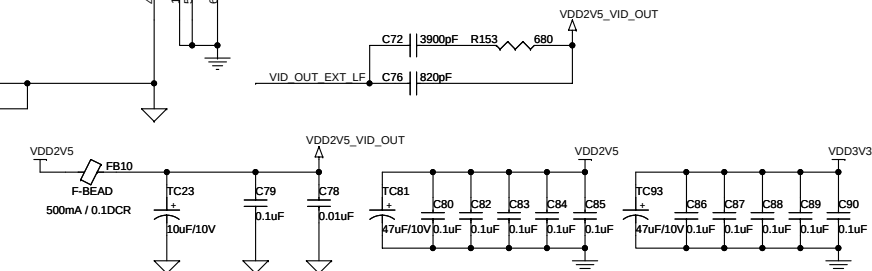


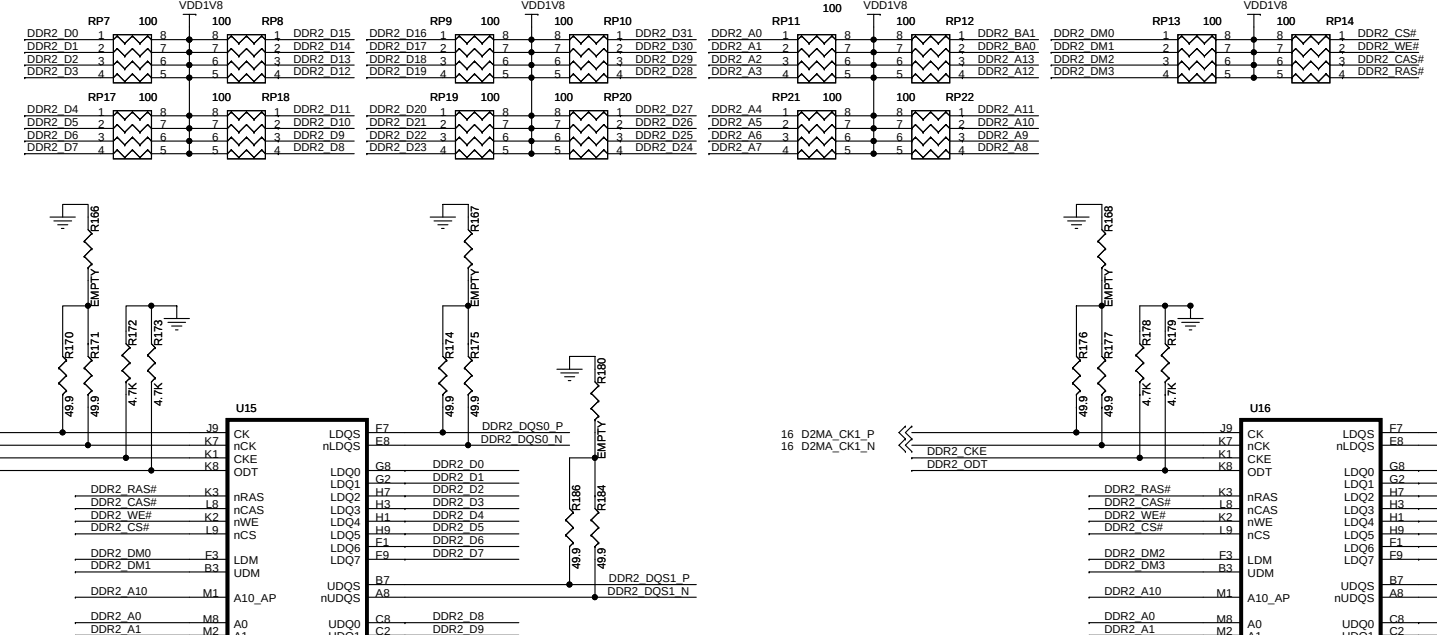
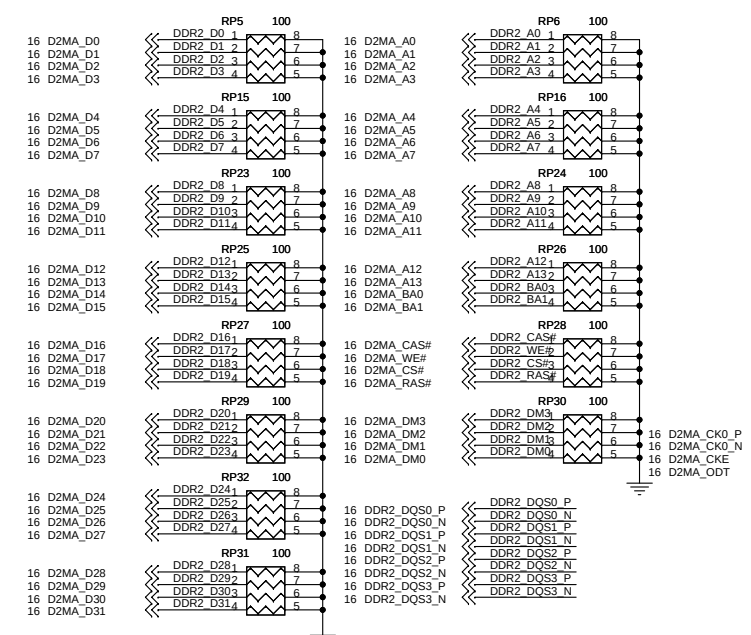
XCCACE_TQ144





VIDEO CONTROL I2C PULL-UP





PLACE CLOSE TO DDR MEMORY DEVICES

NOTE THAT CK & DQS HAVE DIFFERENTIAL TERMINATIONS.

DQ/DQS FOR COMPONENTS DO NOT HAVE ANY TERMINATION AT THE FPGA END.

THE INTENTION IS TO RELY ON SSTL18_II_DCI I/O OF FPGA FOR TERMINATION.

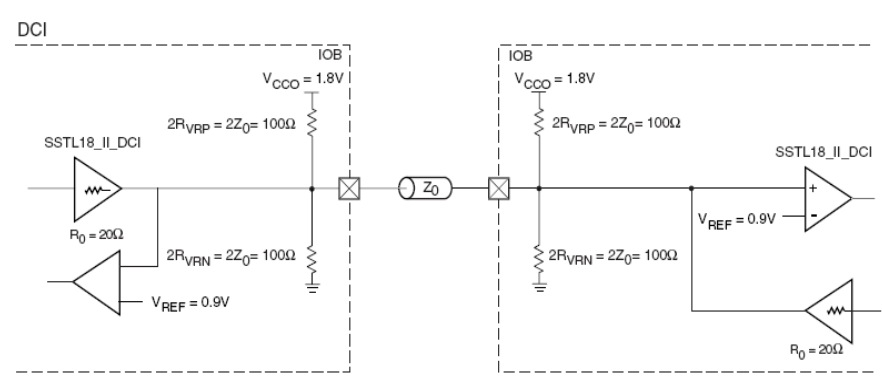
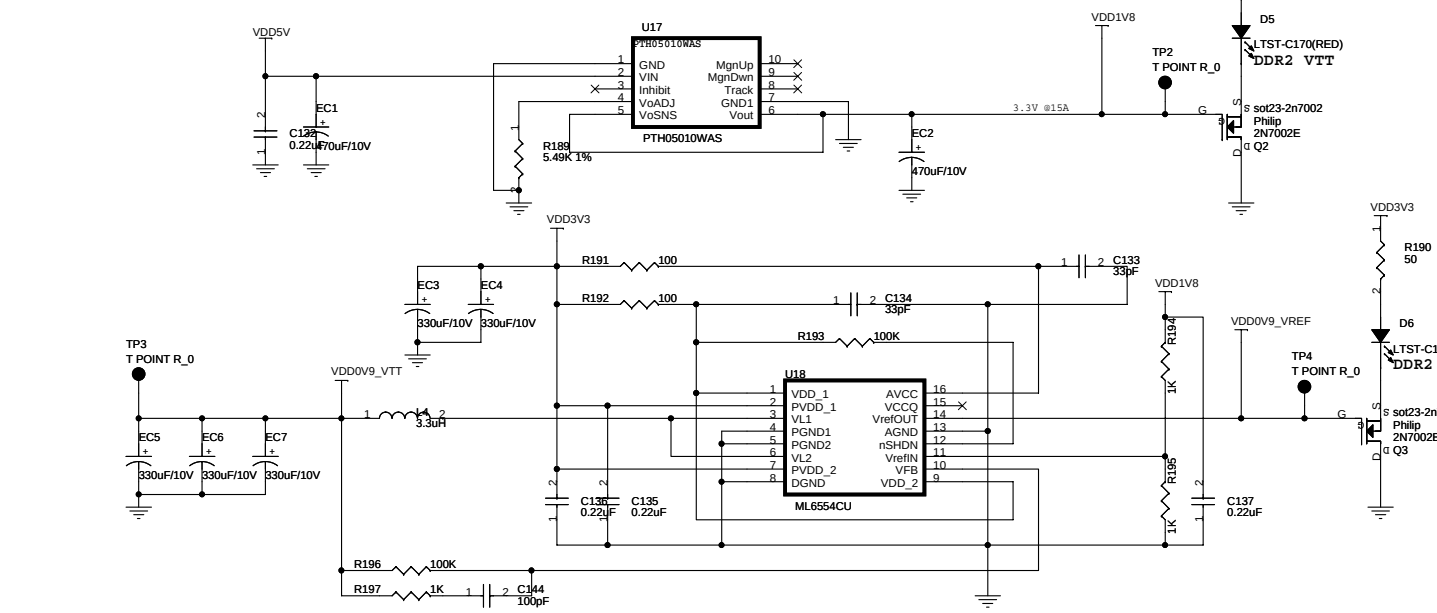
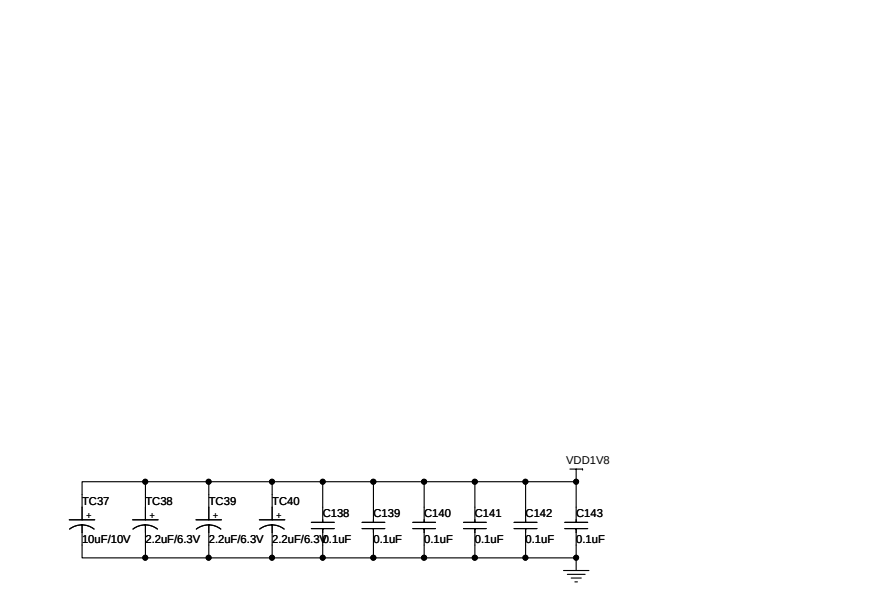
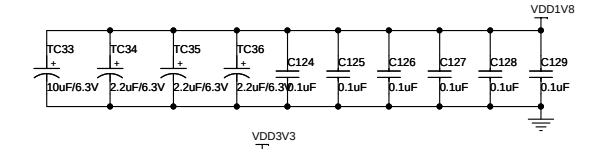
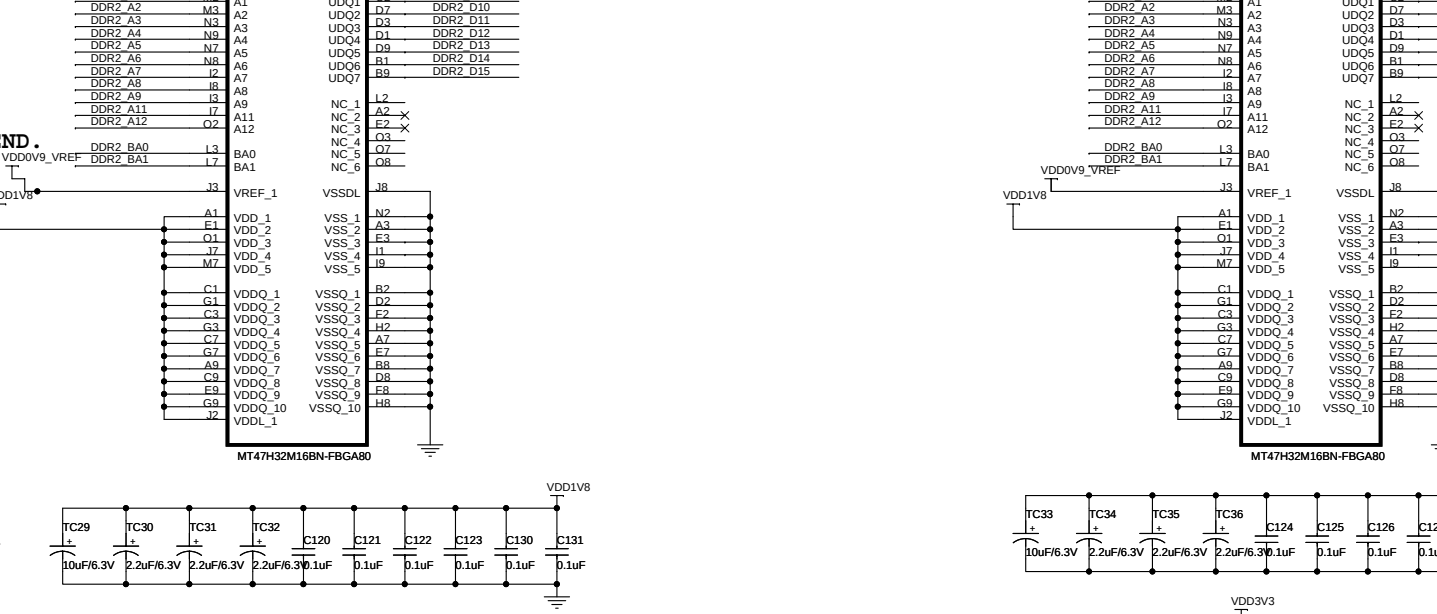
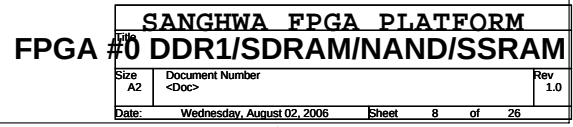
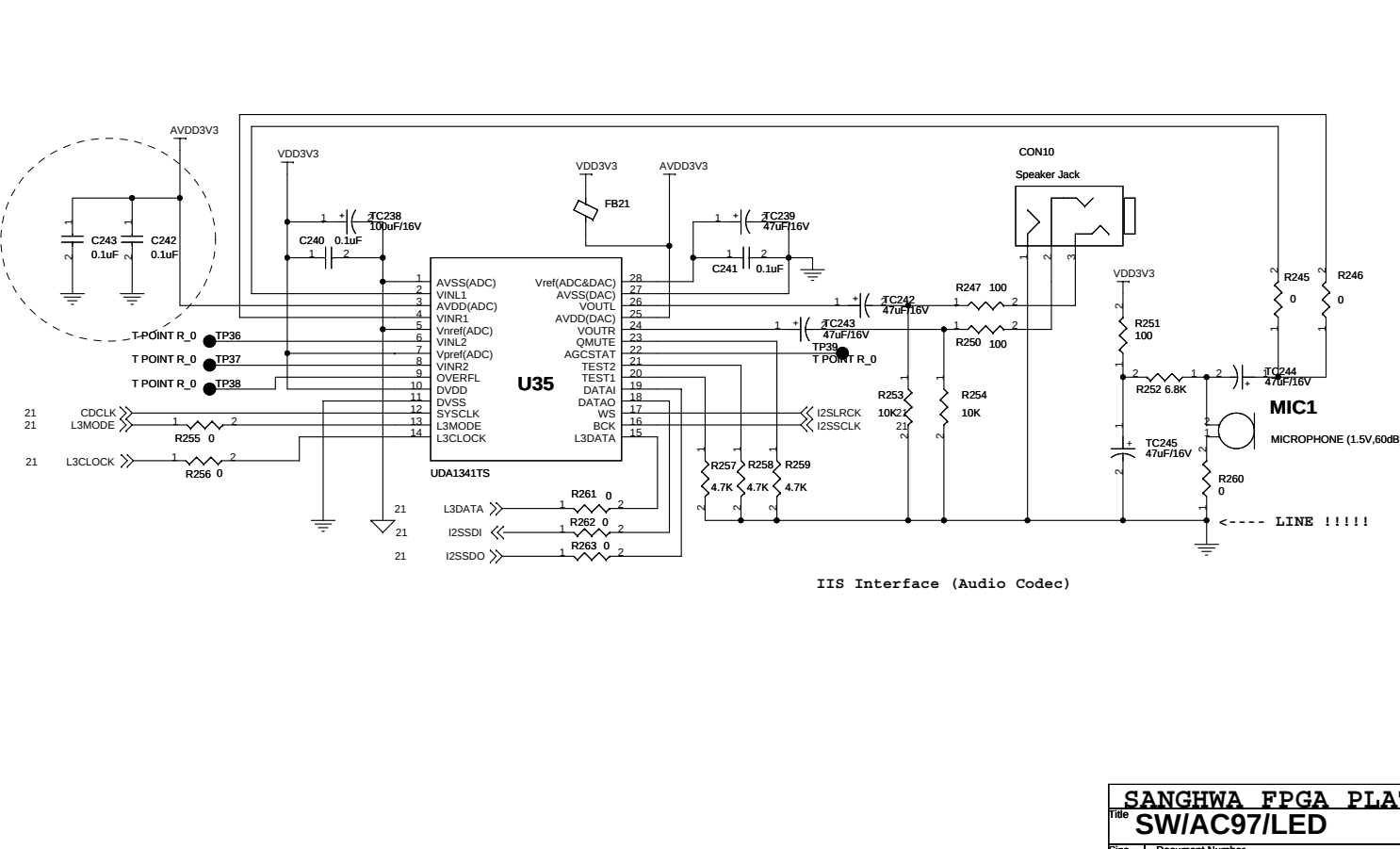
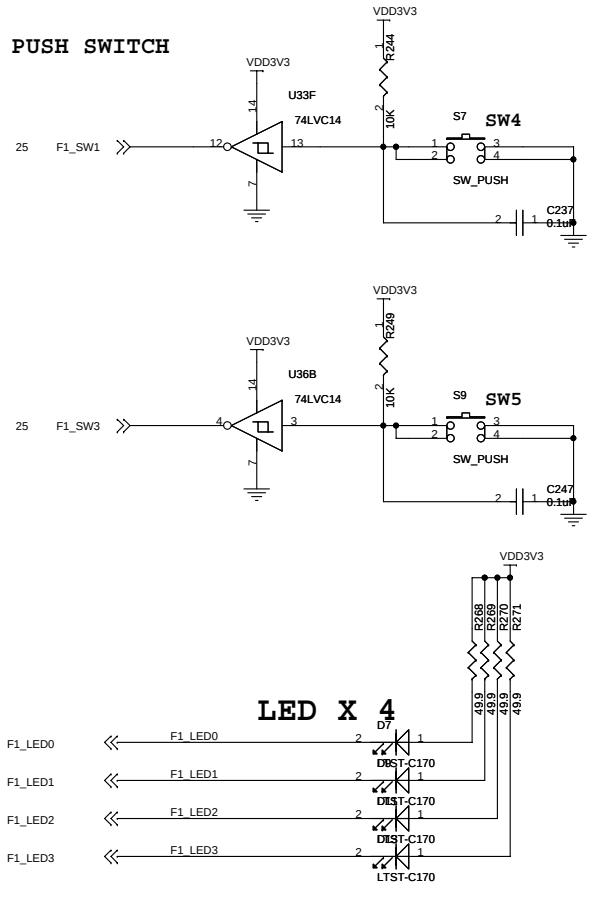
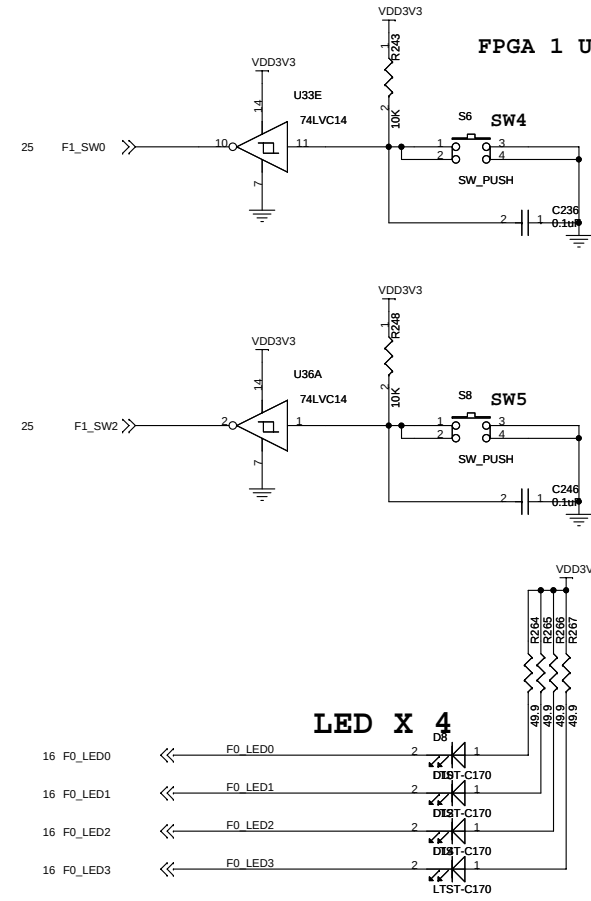
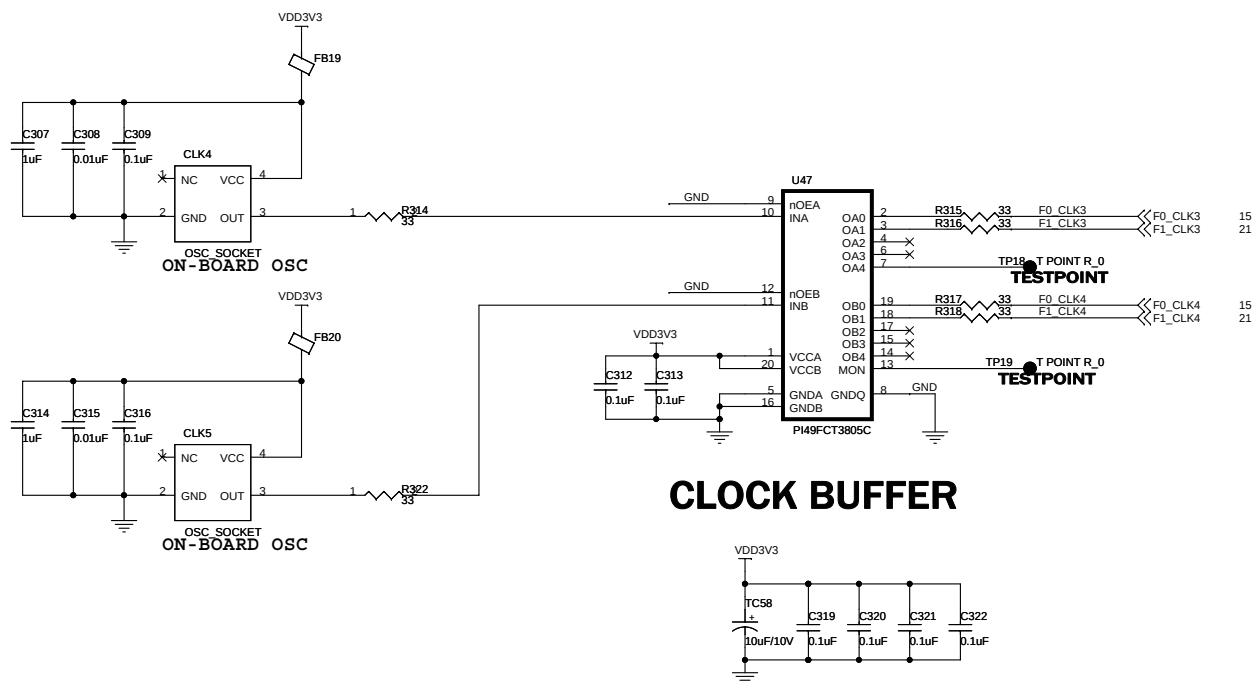
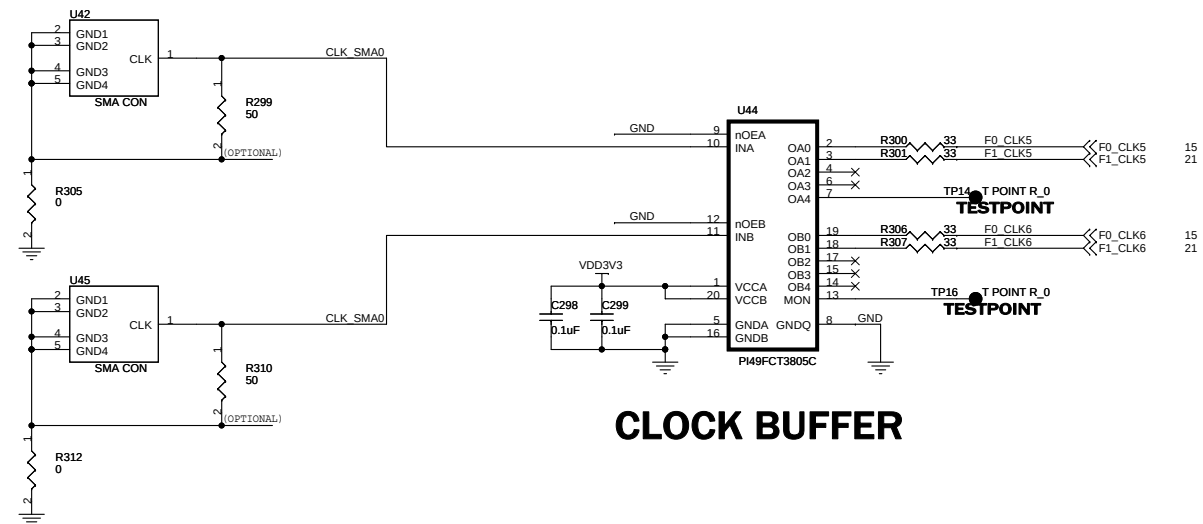
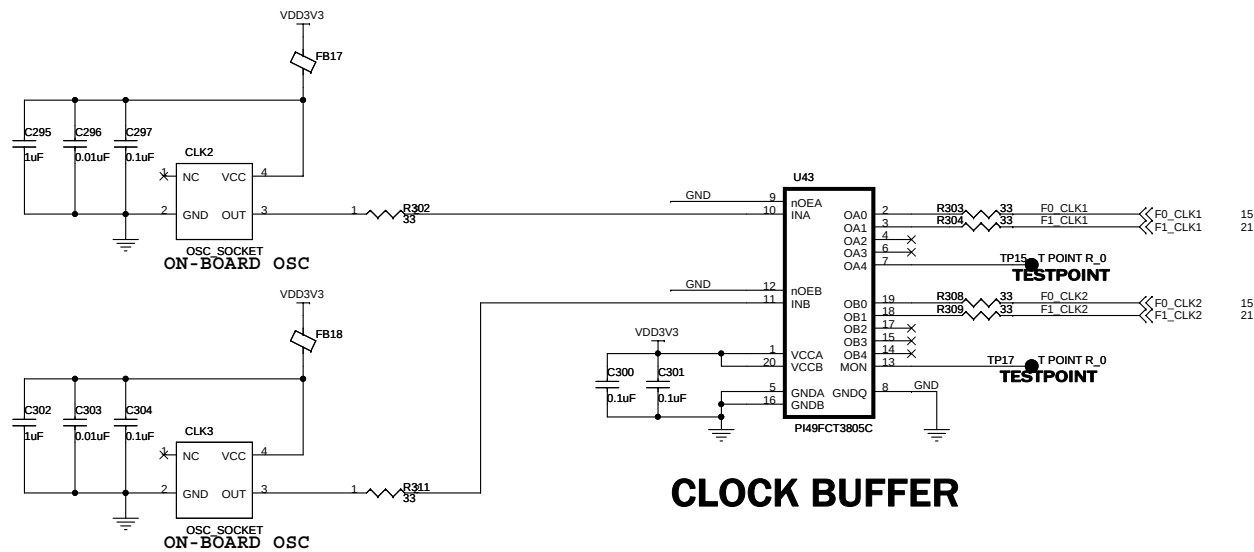


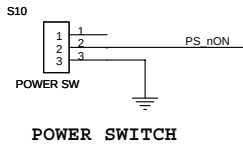
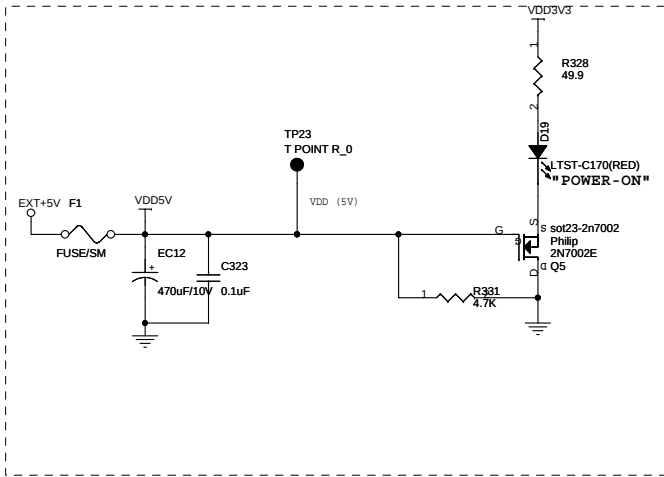
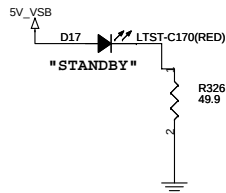
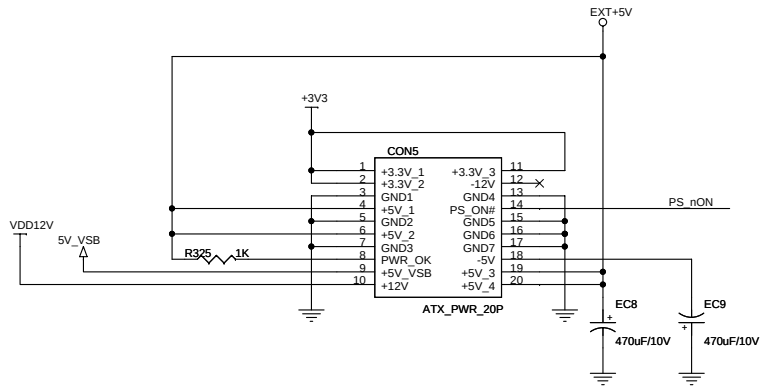
Figure 6-68: SSTL (1.8V) Class II Termination





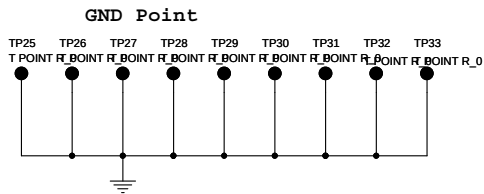
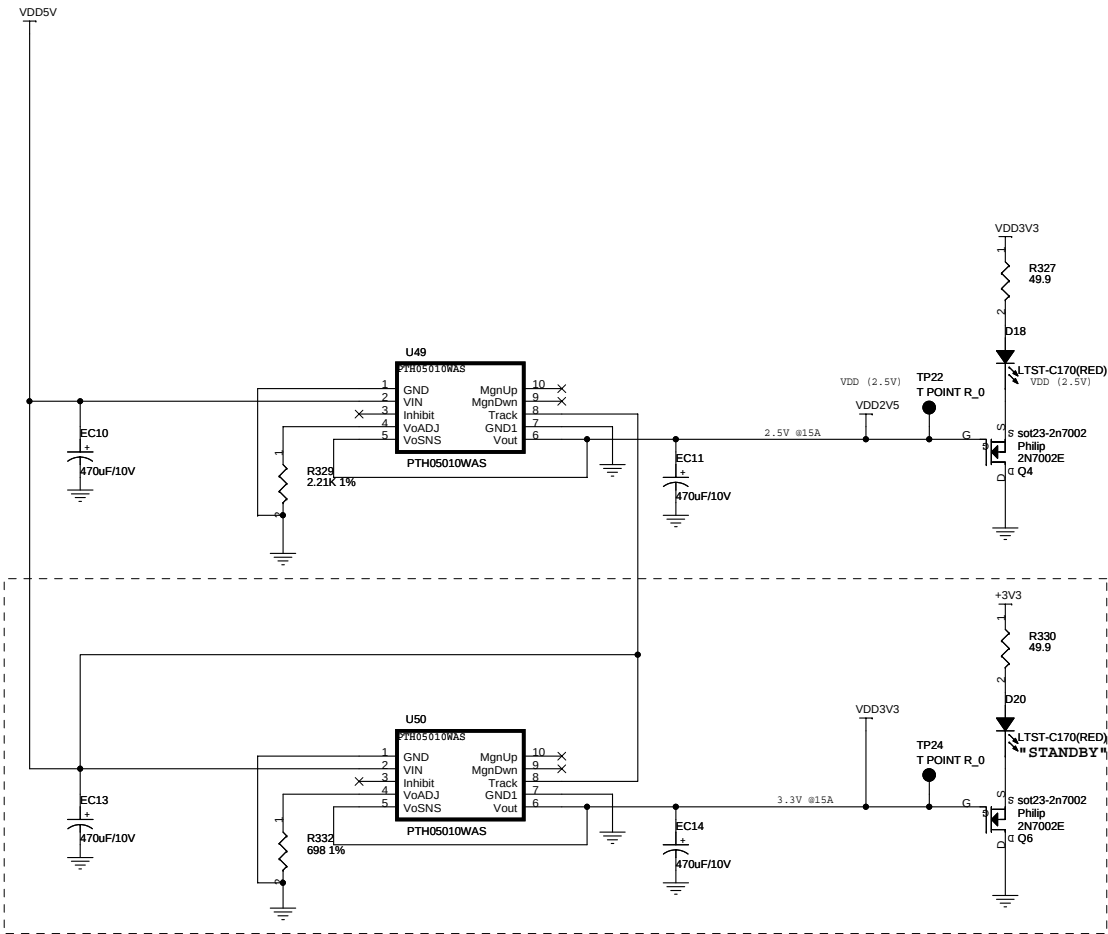




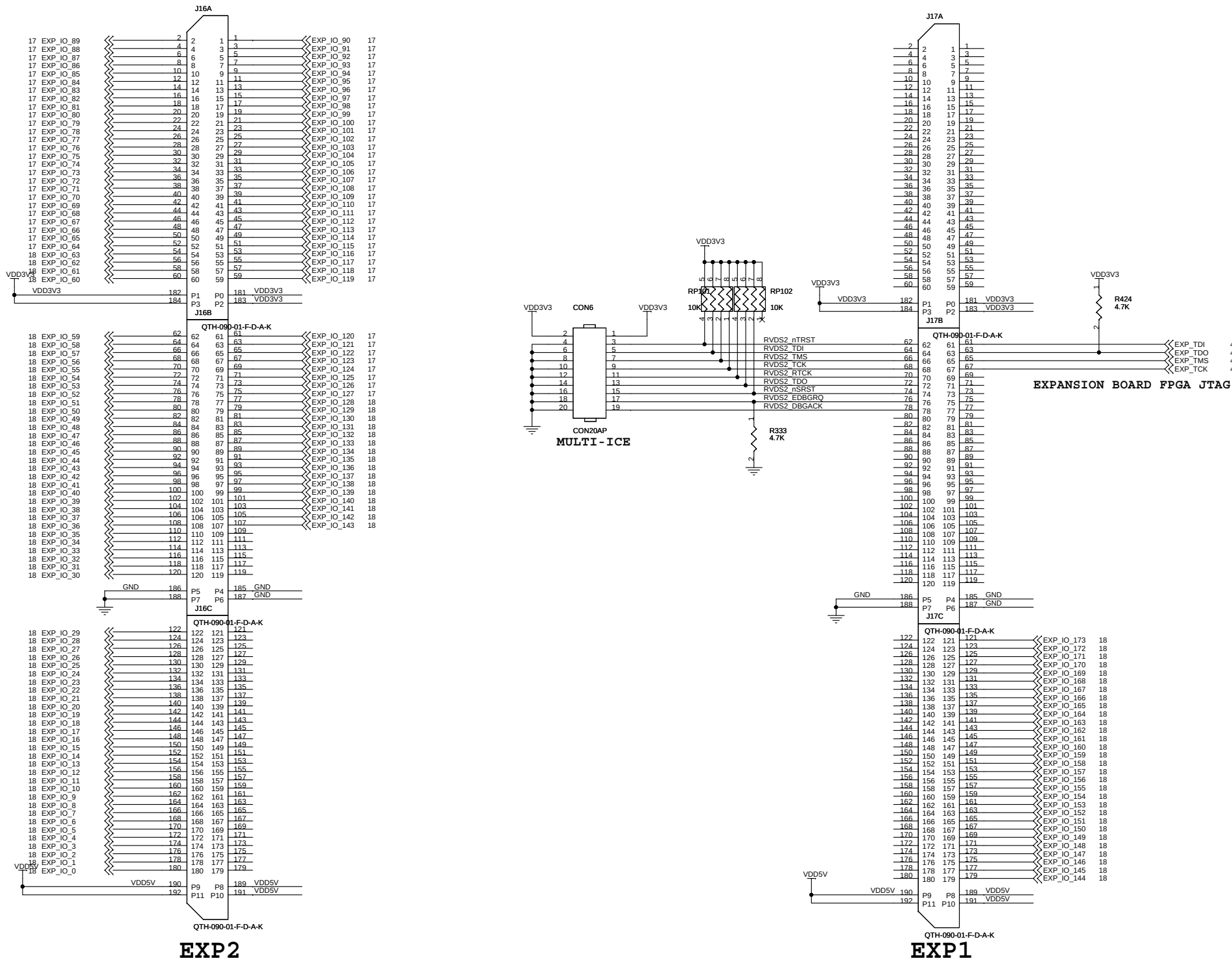


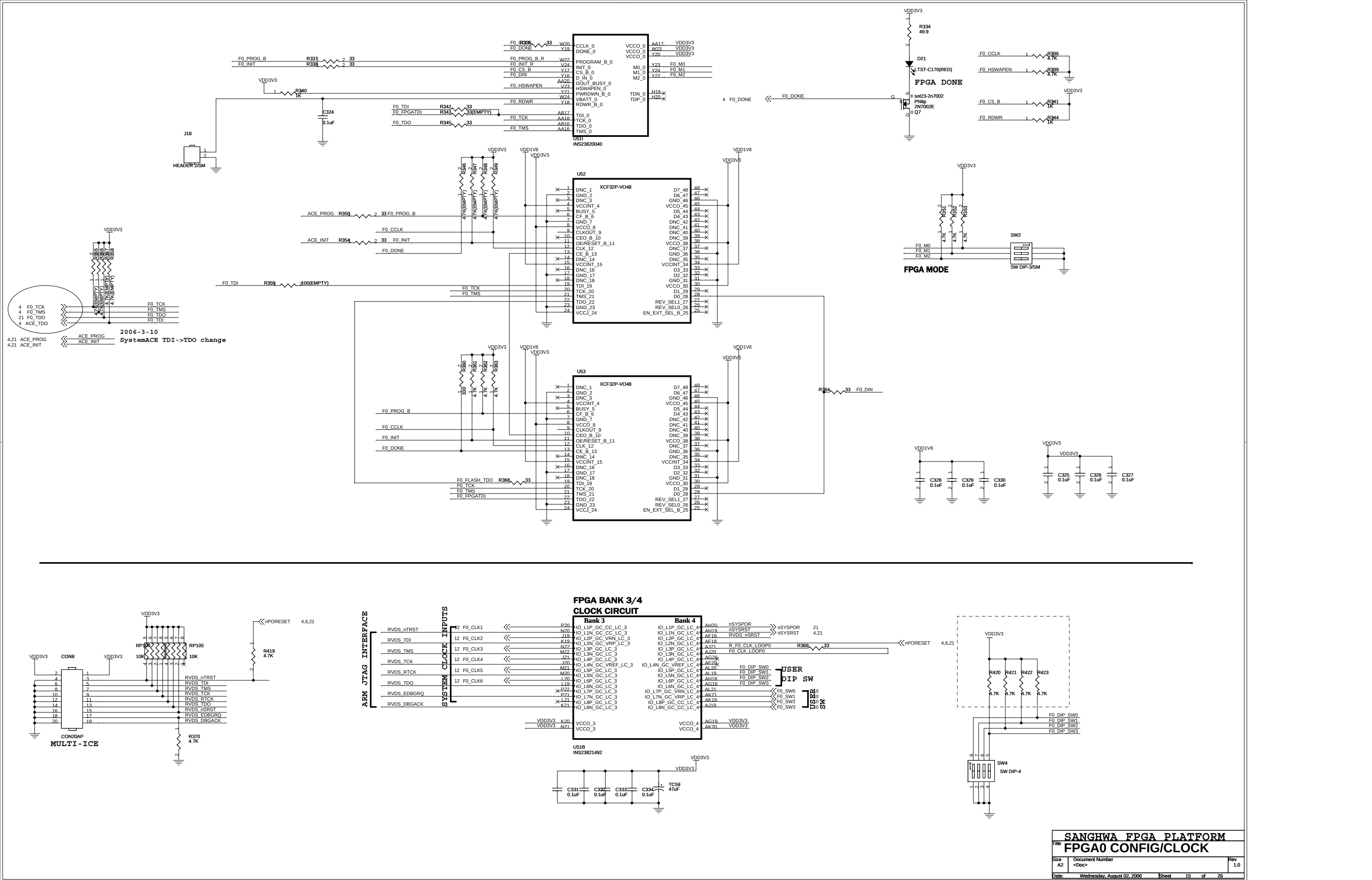
파워스위치 변경

부품 디멘전에 주의 요망!!!!



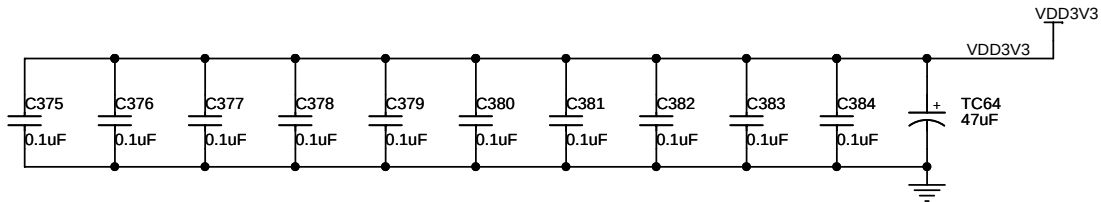
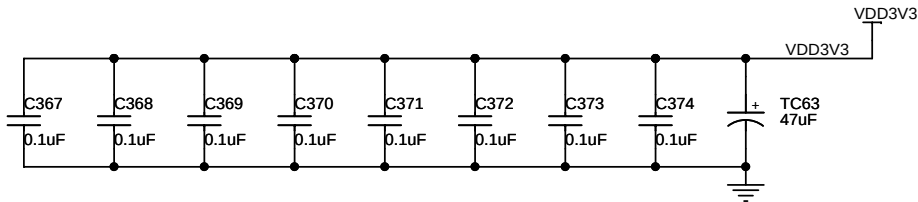
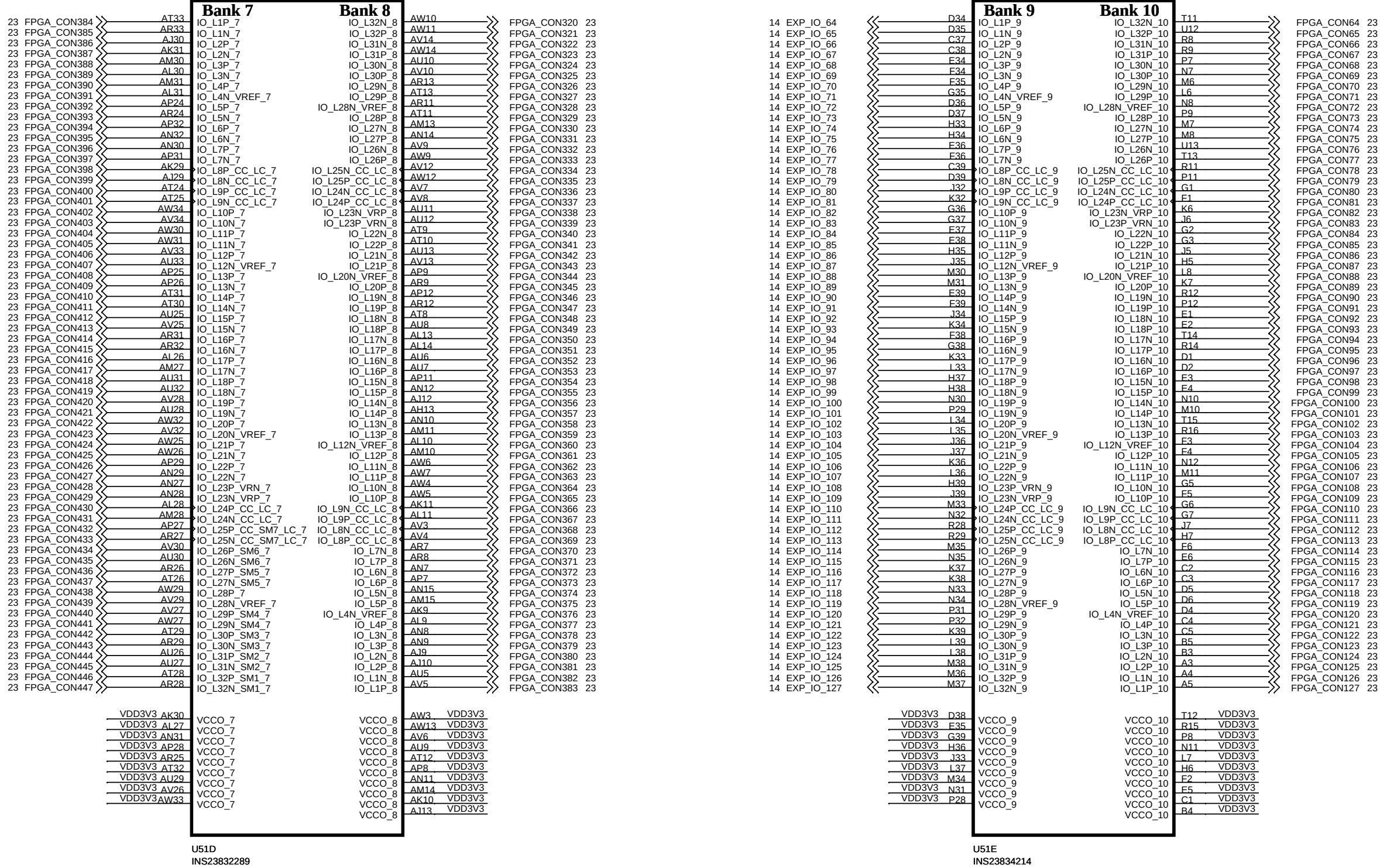
EXPANSION SLOT FOR FPGA #0





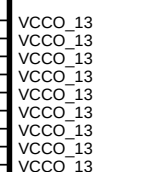
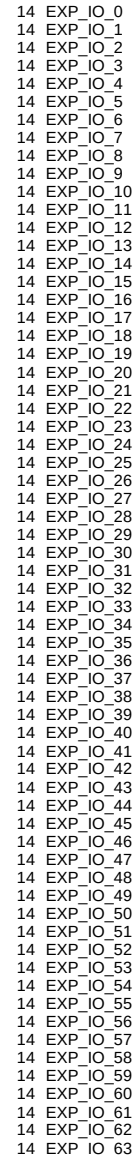
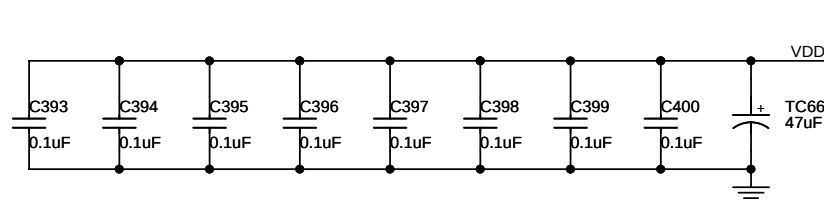
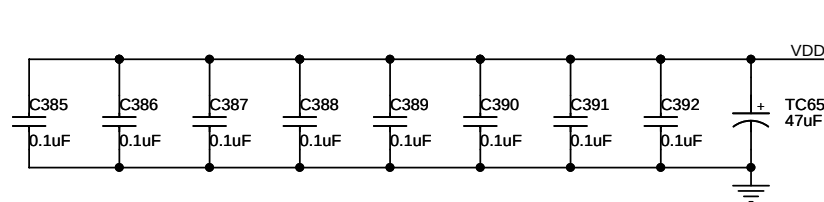
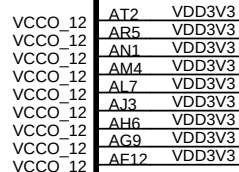
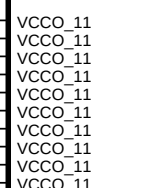
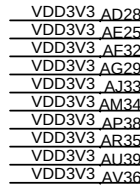
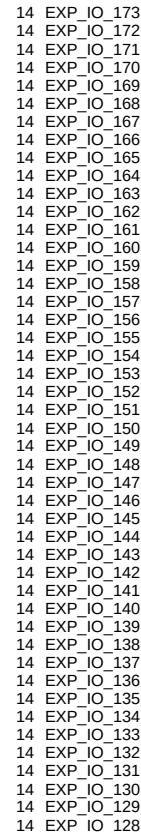


FPGA INTERCONNECTION

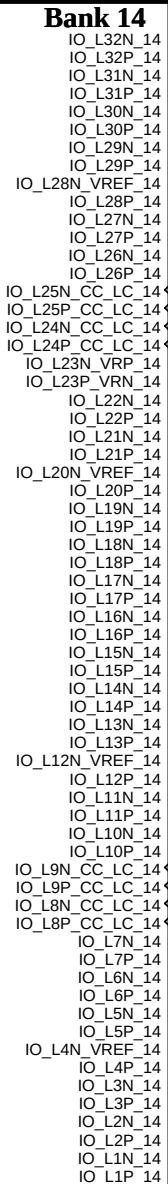



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ON256 24      14 EXP_IO_0
ON257 24      14 EXP_IO_1
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SSRAM INTERFACE



U51G
INS23841562



VIRTEX-4 FPGA

POWER GROUND

GND D13	GND	GND J28	GND
GND K13	GND	GND N28	GND
GND M13	GND	GND W28	GND
GND P13	GND	GND AJ28	GND
GND AD13	GND	GND AW28	GND
GND AF13	GND	GND B29	GND
GND AK13	GND	GND M29	GND
GND AP13	GND	GND AB29	GND
GND G14	GND	GND AM29	GND
GND L14	GND	GND E30	GND
GND N14	GND	GND R30	GND
GND U14	GND	GND AE30	GND
GND W14	GND	GND AR30	GND
GND AE14	GND	GND H31	GND
GND AG14	GND	GND V31	GND
GND AI14	GND	GND AH31	GND
GND AU14	GND	GND AV31	GND
GND K15	GND	GND A32	GND
GND M15	GND	GND I32	GND
GND P15	GND	GND AA32	GND
GND V15	GND	GND AL32	GND
GND Y15	GND	GND D33	GND
GND AD15	GND	GND P33	GND
GND AE15	GND	GND AD33	GND
GND AK15	GND	GND AP33	GND
GND C16	GND	GND G34	GND
GND N16	GND	GND U34	GND
GND U16	GND	GND AG34	GND
GND W16	GND	GND AU34	GND
GND AC16	GND	GND K35	GND
GND AE16	GND	GND Y35	GND
GND AN16	GND	GND AK35	GND
GND F17	GND	GND C36	GND
GND T17	GND	GND N36	GND
GND V17	GND	GND AC36	GND
GND AE17	GND	GND AN36	GND
GND AT17	GND	GND E37	GND
GND J18	GND	GND T37	GND
GND W18	GND	GND AE37	GND
GND AC18	GND	GND AT37	GND
GND A118	GND	GND A38	GND
GND AW18	GND	GND J38	GND
GND B19	GND	GND W38	GND
GND M19	GND	GND AJ38	GND
GND V19	GND	GND AW38	GND
GND AB19	GND	GND R39	GND
GND AD19	GND	GND M39	GND
GND AM19	GND	GND AB39	GND
GND E20	GND	GND AM39	GND
GND R20	GND	GND AV39	GND

GND AE20	GND	GND R1	GND
GND AR20	GND	GND H1	GND
GND H21	GND	GND V1	GND
GND T21	GND	GND AH1	GND
GND V21	GND	GND AV1	GND
GND AB21	GND	GND A2	GND
GND AH21	GND	GND AA2	GND
GND AV21	GND	GND AL2	GND
GND A22	GND	GND AW2	GND
GND L22	GND	GND D3	GND
GND U22	GND	GND P3	GND
GND AA22	GND	GND AD3	GND
GND AL22	GND	GND AP3	GND
GND D23	GND	GND G4	GND
GND P23	GND	GND U4	GND
GND AB23	GND	GND AG4	GND
GND AD23	GND	GND AU4	GND
GND AP23	GND	GND K5	GND
GND G24	GND	GND Y5	GND
GND R24	GND	GND AK5	GND
GND U24	GND	GND C6	GND
GND AA24	GND	GND N6	GND
GND AC24	GND	GND AC6	GND
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GND AU24	GND	GND E7	GND
GND K25	GND	GND T7	GND
GND P25	GND	GND AE7	GND
GND T25	GND	GND AT7	GND
GND Y25	GND	GND J8	GND
GND AB25	GND	GND W8	GND
GND AE25	GND	GND AJ8	GND
GND AH25	GND	GND AW8	GND
GND AK25	GND	GND B9	GND
GND C26	GND	GND M9	GND
GND L26	GND	GND AB9	GND
GND N26	GND	GND AM9	GND
GND R26	GND	GND E10	GND
GND AA26	GND	GND R10	GND
GND AC26	GND	GND AE10	GND
GND AG26	GND	GND AR10	GND
GND AI26	GND	GND H11	GND
GND AN26	GND	GND V11	GND
GND E27	GND	GND AH11	GND
GND K27	GND	GND AV11	GND
GND P27	GND	GND A12	GND
GND T27	GND	GND I12	GND
GND AE27	GND	GND AA12	GND
GND AH27	GND	GND AG12	GND
GND AK27	GND	GND AL12	GND
GND AT27	GND		

U51K
INS23847039

SSRAM INTERFACE

8 SMA_A2	8 SMA_A3	8 SMA_A4	8 SMA_A5	8 SMA_A6	8 SMA_A7	8 SMA_A8	8 SMA_A9	8 SMA_A10	8 SMA_A11	8 SMA_A12	8 SMA_A13	8 SMA_A14	8 SMA_A15	8 SMA_A16	8 SMA_A17	8 SMA_A18	8 SMA_A19	8 SMA_A20	8 SMA_nADSC	8 SMA_nADSP	8 SMA_nADV	8 SMA_nBW1	8 SMA_nBW2	8 SMA_nBW3	8 SMA_nBW4	8 SMA_nBWE	8 SMA_nCE	8 SMA_CLK	8 SMA_nCS1	8 SMA_nGW	8 SMA_nLBO	8 SMA_D31	8 SMA_D30	8 SMA_D29	8 SMA_D28	8 SMA_D27	8 SMA_D26	8 SMA_D25	8 SMA_D24	8 SMA_D23	8 SMA_D22	8 SMA_D21	8 SMA_D19	8 SMA_D18	8 SMA_D17	8 SMA_D16	8 SMA_D15	8 SMA_D14	8 SMA_D13	8 SMA_D12	8 SMA_D11	8 SMA_D10	8 SMA_D9	8 SMA_D8	8 SMA_D7	8 SMA_D6	8 SMA_D5	8 SMA_D4	8 SMA_D3	8 SMA_D2	8 SMA_D1	8 SMA_D0
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VDD3V3	AA27
VDD3V3	AB34
VDD3V3	AC31
VDD3V3	AD38
VDD3V3	AE35
VDD3V3	AG39
VDD3V3	AH36
VDD3V3	AI37
VDD3V3	Y30

Bank 15

IO_L1P_15	IO_L1N_15	IO_L2P_15	IO_L2N_15	IO_L3P_15	IO_L3N_15	IO_L4P_15	IO_L4N_VREF_15	IO_L5P_15	IO_L5N_15	IO_L6P_15	IO_L6N_15	IO_L7P_15	IO_L7N_15	IO_L8P_CC_LC_15	IO_L8N_CC_LC_15	IO_L9P_CC_LC_15	IO_L9N_CC_LC_15	IO_L10P_15	IO_L10N_15	IO_L11P_15	IO_L11N_15	IO_L12P_15	IO_L12N_VREF_15	IO_L13P_15	IO_L13N_15	IO_L14P_15	IO_L14N_15	IO_L15P_15	IO_L15N_15	IO_L16P_15	IO_L16N_15	IO_L17P_15	IO_L17N_15	IO_L18P_15	IO_L18N_15	IO_L19P_15	IO_L19N_15	IO_L20P_15	IO_L20N_VREF_15	IO_L21P_15	IO_L21N_15	IO_L22P_15	IO_L22N_15	IO_L23P_VRN_15	IO_L23N_VRP_15	IO_L24P_CC_LC_15	IO_L24N_CC_LC_15	IO_L25P_CC_LC_15	IO_L25N_CC_LC_15	IO_L26P_15	IO_L26N_15	IO_L27P_15	IO_L27N_15	IO_L28P_15	IO_L28N_VREF_15	IO_L29P_15	IO_L29N_15	IO_L30P_15	IO_L30N_15	IO_L31P_15	IO_L31N_15	IO_L32P_15	IO_L32N_15
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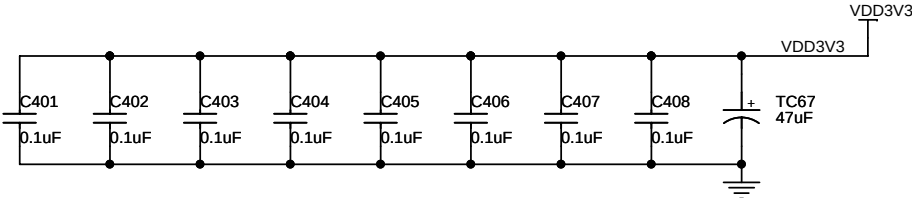
U51H
INS28868288

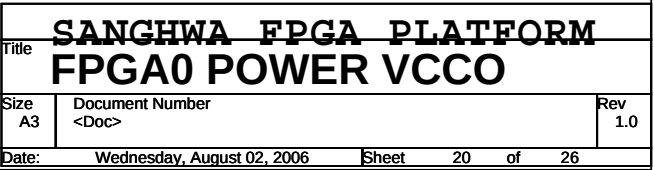
Bank 16

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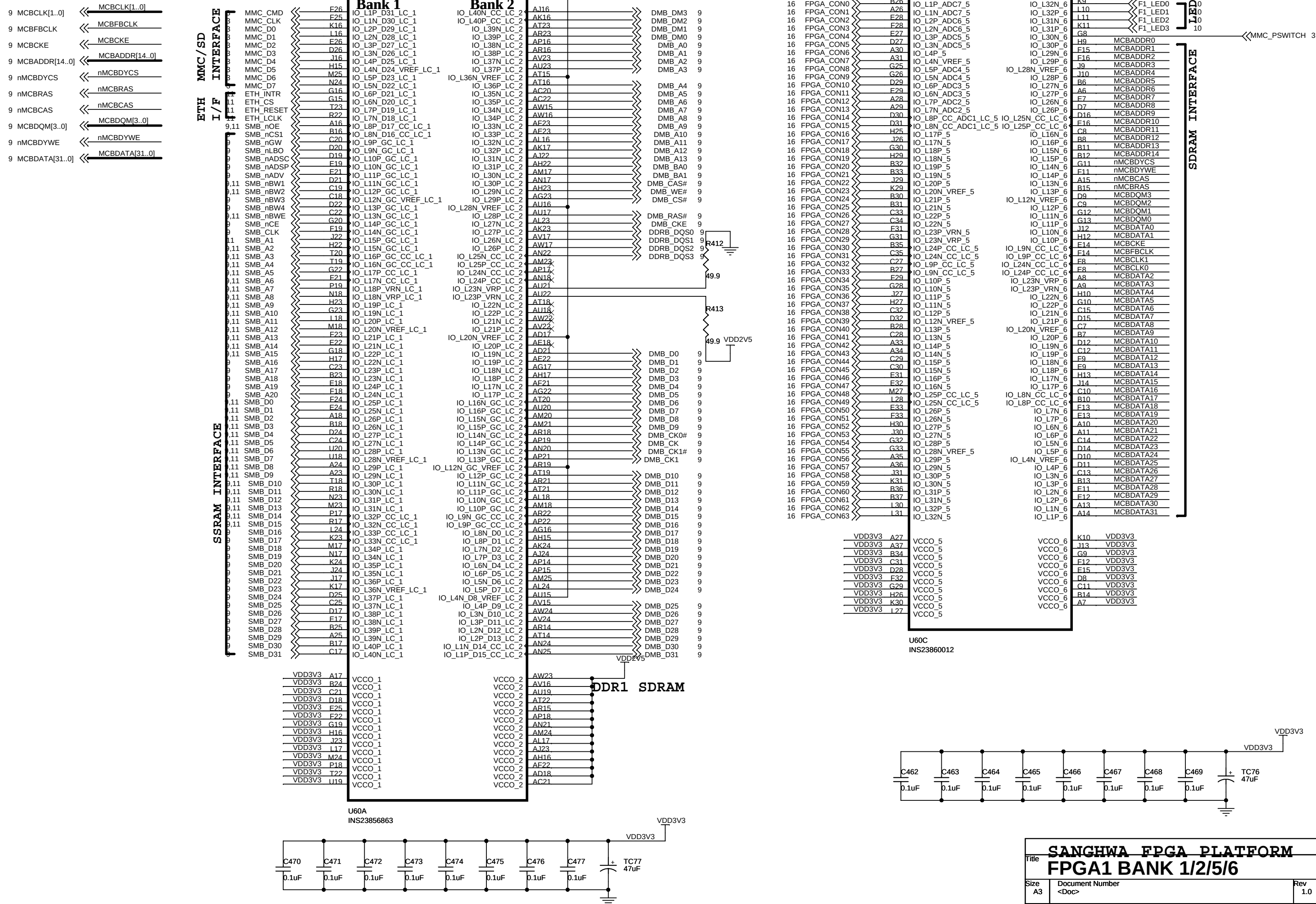
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AE2	VDD3V3
AE5	VDD3V3
AD8	VDD3V3
AC11	VDD3V3
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AB4	VDD3V3
AB14	VDD3V3
AA7	VDD3V3

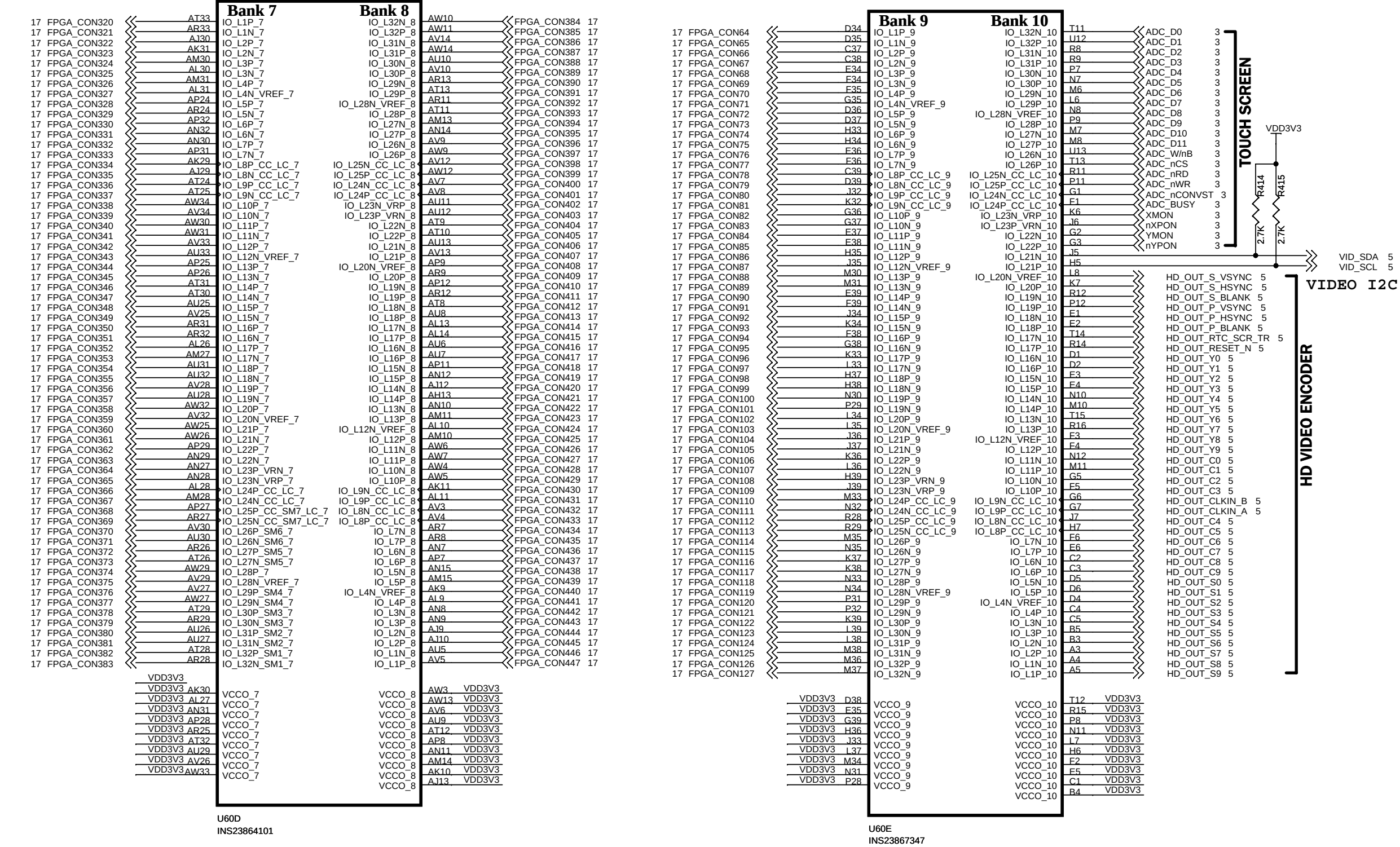
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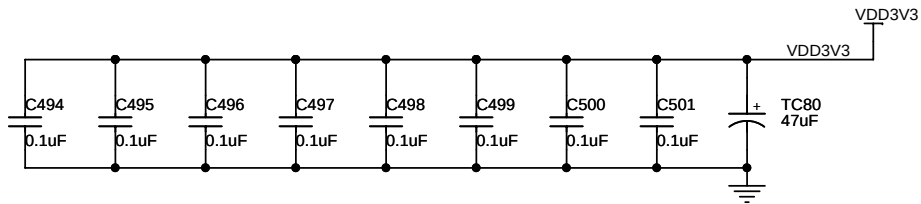
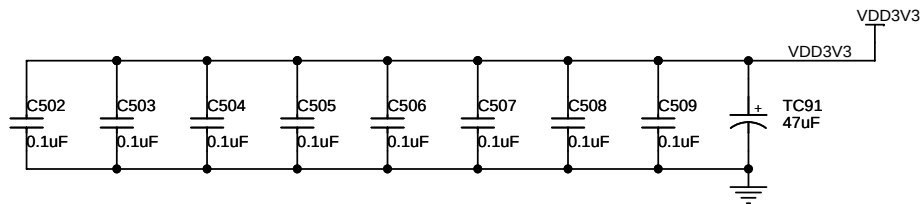
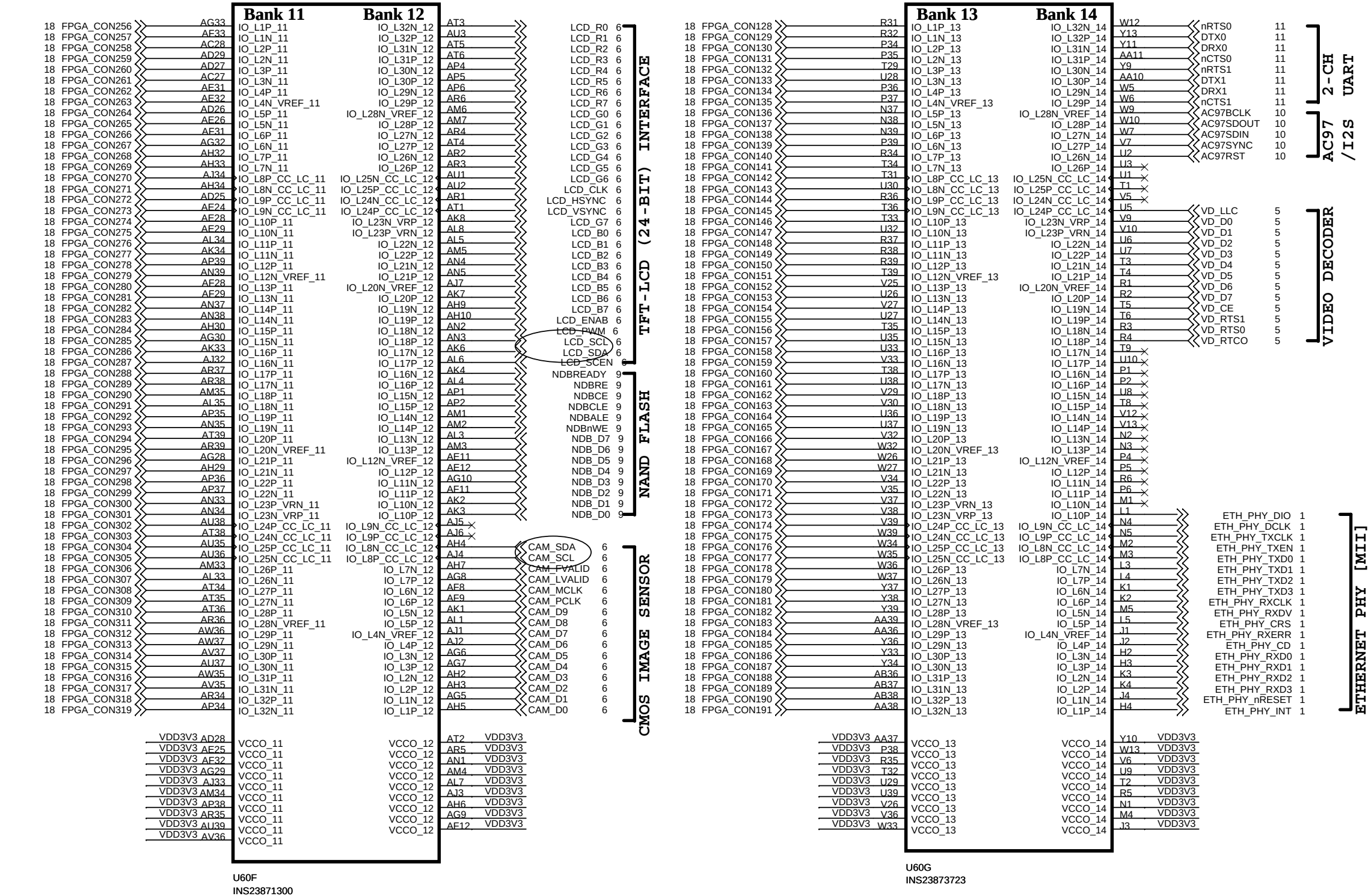




DYNAMIC MEMORY INTERFACE







**VIRTEX-4 FPGA
POWER GROUND**

